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EDUCATION

University of North Carolina at Chapel Hill (UNC-CH) | Chapel Hill, NC

M.S. in **Information Science (MSIS)** *Jan 2026 - Exp. May 2027*

School of Information and Library Science (SILS). BS-MS Dual Degree Program.

B.S. in **Information Science** and **Neuroscience**, Minor in **Computer Science** *Aug 2022 - Dec 2025*

Relevant Coursework: Human Information Interactions, Open-Source (OSINT), Health Informatics, AI Literacy & Law.

North Carolina School of Science and Mathematics (NCSSM) | Durham, NC *Aug 2020 - May 2022*

TECHNICAL SKILLS

Programming: SQL, Python, HTML/CSS, Java, JavaScript, React, C, Assembly, Dart.
Frameworks & Libraries: FHIR, TensorFlow/Keras, MNE (EEG), Flutter, Git, MATLAB, Node.js, Three.js, Leaflet, AJAX.
Software: Blender, GitHub, Onshape, Twine, Figma, Wix, WordPress, Schrödinger Maestro, Gaussian, ChimeraX, WebMO.
Tools: Microsoft Office (Excel, Word, PowerPoint), Google Suite, Adobe Photoshop/Acrobat, TinkerCAD, AutoCAD, Notion.
Languages: English (Fluent), Bengali (Fluent), Spanish (Proficient), Hindi (Novice).

RESEARCH PROJECTS

Eye-Tracking Engagement in AI-Assisted Academic Search *Chapel Hill, NC | Sept 2025 - Present*

- Designed a mixed-methods evaluation comparing search behavior with vs. without AI support, measuring engagement, efficiency, and comprehension during research tasks.
- Integrated eye-tracking via **Tobii Nano Pro.** Collected and processed eye-movement data, pairing gaze signals with behavioral metrics to understand visual attention patterns in adaptive learning tools and identify friction points.
- Translating findings into actionable interface design improvements and informing AI guidelines in education.

LLM-Driven Language Shift in Naturalistic Communication *Chapel Hill, NC | Aug 2025 - Feb 2026*

- Executed a **PRISMA**-guided systematic review across 6 databases & 4 preprint servers, synthesizing and evaluating evidence for LLM-associated changes in human language and communication patterns.
- Developed a structured **five-lens conceptual extraction framework** to see how studies operationalize LLM exposure (direct use vs. societal diffusion) and measure linguistic outcomes, surfacing gaps and methodological weaknesses.

Brain-Computer Interface (BCI) Model Card – DREAMS Framework Evaluation *Chapel Hill, NC | Jul 2025 - Dec 2025*

- Built end-to-end BCI pipeline in TensorFlow/Keras/MNE, applying ML models to neural signal data. Trained and benchmarked multiple **CNN** architectures (**EEGNet, DeepConvNet, & ShallowConvNet**), generating performance metrics, confusion matrices, and uncertainty estimates.
- Evaluated DREAMS HTML model cards with a 13-item **System Explainability Scale (SES**) across Trust, Usability, and Understandability/Interpretability dimensions. Documented framework components (preprocessing, metrics, hyperparameters) and limitations under real-world noisy EEG conditions. Published reproducible artifact to GitHub.

Nanotechnology Treatment for Cancer, Research & Innovation Program (SRIP) *Durham, NC | Apr 2021 - May 2022*

- Self-synthesized biogenic silver nanoparticles using agri-food byproducts as a novel anticancer reducing agent.
- Employed computational chemistry molecular modeling techniques with **Schrödinger Maestro**, **Gaussian**, **MATLAB**, **Vortex**, and **WebMO**, alongside wet lab research for physicochemical property analyses and medical drug design. **3D-printed** ligands, modelled in **ChimeraX**, to illustrate orthostatic binding mechanisms.
- One of the **Top 300** Regeneron Science Talent Search (STS) Scholars selected out of **2,162** applicants internationally. Winner of NC International Science Challenge issued by the NC SMT Center and Morehead Planetarium.

PROGRAMMING PROJECTS

Health Chart App – Electronic Health Record (EHR) Data Visualization *Chapel Hill, NC | Sept 2025 - Dec 2025*

- Built a JavaScript web app (**Leaflet.js**, **AJAX**) as an EHR dashboard and interface to visualize patient vitals and trends.
- Implemented **FHIR (Fast Healthcare Interoperability Resources) API** for real-time data integration with hospital systems like **EPIC**. Proposed application extensions to support clinical decision-making and patient communication.

3D-Printed Receptor Model – Advanced Molecular Neuropharmacology. *Chapel Hill, NC | Aug 2025 - Dec 2025*

- Designed and fabricated a functional, interactive physical model of the D3 dopamine receptor (GPCR) demonstrating conformational change triggered by a bitopic ligand (FOB02-04A), using cryo-EM structures.

- Processed receptor structures in **ChimeraX** and **Blender**; performed CAD modeling in **TinkerCAD** and **Onshape** to carve binding pockets; 3D printed in **PLA/PETG** on Makerspace FDM printers through multiple design iterations.
- Presented working model and research poster at UNC BeAM Makerfest as a communication tool for medicinal chemists and clinicians to visualize D3-selective bitopic agonist activity relevant to Parkinson's disease therapeutics.

Digital Art Exhibition Catalog - New Media Course Final *Chapel Hill, NC | Jun 2024 - July 2024*

- Created immersive virtual gallery using JavaScript **Three.js** 3D library and **Blender**.
- Integrated narrative **Twine** scripting and layered media transitions with **Photoshop** for interactive storytelling.

Temperament Test Website - Pearl Hacks Hackathon *Chapel Hill, NC | Feb 2023*

- Developed full-stack web app dynamic JavaScript based front-end for a personality quiz that generated temperament profiles through user-interactive forms. Local storage and responsive interface design for accessibility.
- Awarded **Highest STEM-Interest Inclusivity Runner-Up** by Qorvo Inc.

LEADERSHIP EXPERIENCE & COMMUNITY SERVICE

Facilitator & College Loop President — Girls Who Code, UNC Chapter *Chapel Hill, NC | Jan 2023 - Present*

- Developed lesson plans, built activities in **Turtle**, and taught **Python** programming to **50+** students weekly. Lectured and held office hours; mentored project development; led instruction through semester-long app design projects.
- Founded UNC's GWC College Loop (2026); guided coding projects and partnered with local schools to expand CS access for underrepresented groups; communicated with parents, alumni, and sponsors.

Ambassador & Student Advisory Board Member — UNC School of Information and Library Science (SILS) *Chapel Hill, NC | Aug 2025 - Present*

- Selected to represent SILS for prospective students and families through outreach events and academic panels.
- Serve on the **SILS New School Student Advisory Board**. Provide input to school leadership on curriculum, student experience, spaces, and resources during the SILS-SDSS merger. Advise institutional decision-making.

Co-President — UNC Zero Degrees *Chapel Hill, NC | Mar 2025 - Present*

- Scaled UNC's largest interdisciplinary social networking organization to **200+** members on HeelLife and Discord; secured **$1800** in Undergraduate Senate Funding for programming and community events.
- Managed a platform for student communication. Implemented automated moderation tools, interactive bots, and event analytics, expanding student engagement by **85%**.

Co-Founder & Chief Operating Officer — SkyeLabs Innovation Inc. *Durham, NC | Sept 2020 - Aug 2024*

- Startup on sustainable materials research. Directed product development, business plans, and organizational operations for **EcoGel**, a sustainable biopolymer aerogel for infrastructure and high-performance apparel.
- Led team through R&D, funding strategy, and prototyping phases. Filed provisional patent. Secured **$2000** in research grant funding from the Bowman-Brockman Endowment for Entrepreneurship and Advanced Research.
- Won **Pete Conrad Scholar** (International Conrad Spirit of Innovation Challenge 2021, Aerospace & Aviation); recognized as **Top 10 Finalist** at **NASA iTech** & sole Honorable Mention as the youngest team recognized nationally.

Apprentice — Girls Who Code × Pluralsight Work Prep Apprenticeship *Remote | May 2023 - Jun 2023*

- Selected for Pluralsight workshops in full-stack web & software development, Python, Java, and cloud technologies.
- Gained experience with **SDLC**, **Agile** & **Waterfall** methodologies, **Git**, **DevOps**, and **CI/CD** pipelines.
- Handpicked as Closing Ceremony speaker out of **100+** participants; moderated official Discord server for students.

Editor-in-Chief — Broad Street Scientific (BSS) *Durham, NC | Oct 2020 - Jun 2022*

- Oversaw annual **100-page** research publication and mentored editorial staff of **20+** student editors and writers.
- Organized and facilitated student publishing workshops. Designed submission management system for the journal website, employing **SEO Optimization**. Critiqued and reviewed student submissions.
- Published NCSSM-Durham's official journal of student STEM research.

HONORS & AWARDS

Pearl Hacks Hackathon Award (2023); Regeneron Science Talent Search Scholar (2022, top 300 of 2,162 applicants); National JSHS Momentum Grant (2022); NC (North Carolina) International Science Challenge Winner (2022); Bowman-Brockman Endowment Recipient (2022); NASA iTech Honorable Mention & Top 10 Finalist (2021); Pete Conrad Scholar (2021); Coca-Cola Scholar Semifinalist (2021, top 2% of 68,000+ applicants); Carson Scholar (2021); AJAS (American Junior Academy of Sciences) Fellow & Lifetime Honors (2020-2021); Intel ISEF (International Science & Engineering Fair) Finalist (2019); National Speech & Debate Academic All-American; World Science Scholar (2019); USCF Top 34 Female Player in Age Group (2019).